# Jesús M. Rueda-Becerril | Ph.D.

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## Professional Summary

Research engineer and computational scientist with a PhD in Physics and 10+ years of experience building physics-based numerical models (PDE solvers, radiative transfer, MHD) and applying machine learning to physical-science problems. Python-fluent, with GPU and HPC experience and an open-source track record: architected and maintained simulation libraries adopted by other research groups, tested and shipped code through CI/CD with test suites, and mentored students. Strong at pairing physical principles with data-driven models and explaining the results to scientific and engineering audiences.

## Skills

**Languages**: Python, C/C++, Fortran 95, Rust, R, Java, SQL, Julia, Shell

**ML & Data**: Scientific ML, hybrid physics–ML modeling, PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, Pandas, Astropy, ML-assisted parameter estimation, gradient-based optimization, SVM, random forest, neural networks

**Geospatial & Remote Sensing**: Sentinel-1/2 multispectral processing, raster/vector data pipelines, QGIS, geospatial feature engineering for ML

**HPC & Parallel**: OpenMP, MPI, OpenACC, SLURM, PBS, HDF5, large-scale parallel CPU computing

**Simulation**: Physics simulation (multiscale/multiphysics modeling, radiative transfer, relativistic MHD, Lagrangian particle dynamics), numerical PDEs (advection–diffusion, Fokker–Planck), finite difference/volume methods, numerical linear algebra, synthetic dataset generation, uncertainty quantification, parameter estimation

**Tools**: Git and code review, Docker, Azure, CI/CD (Jenkins), Kafka, Jira, QGIS, ParaView, VisIt

## Experience

**Independent Contractor** **Jan 2026 – Present**
*Mercor* *Remote*
- Physics expert for a leading AI lab: reviewed 50+ expert-level problems in physics, hydrodynamics, astrophysics, and cosmology; wrote original prompts that stumped frontier models.
- Diagnose failure modes in model reasoning and write structured feedback to improve scientific accuracy.

**Spatial Data Scientist** **May 2025 – Jan 2026**
*TealWaters* *Seattle, WA*
Built geospatial ML pipelines and optimized terrain-analysis code for wetland mapping.
- Mapped wetland probability across the 2,160 km² Skykomish watershed with the Wetland Intrinsic Potential (WIP) random-forest model, reaching >75% accuracy.
- Optimized the terrain-modeling code that computes elevation derivatives from 1 m, 3 m, and 10 m DEMs over 20 tiles, improving its computational performance.
- Engineered features from Sentinel-1/2 multispectral imagery and terrain derivatives for ML land-cover and wetland models; built Python tools and QGIS visualizations.
- Presented results to managers and decision-makers, trained 2 colleagues on the WIP tool, and worked with software engineers to move prototype tools into production.

**Independent Researcher & Open-Source Developer** **Jan 2024 – May 2025**
*Self-directed* *Seattle, WA*
Continued simulation research and open-source development between industry roles.
- Co-developed Tleco, an open-source Rust/Python toolkit that models radiation from relativistic outflows by solving the Fokker–Planck equation coupled to radiative transfer; published in ApJ (2024) [9].
- Developing WindsOfChange with the UW Forest Mycobiome Lab: a C++/R library that simulates wind-driven spore dispersal over forested, hilly terrain, parallelized with RcppParallel and OpenMP.
- Invited talk on machine learning in environmental science at the University of Washington (2024).

**Software Engineer** **Apr 2022 – Jan 2024**
*Paychex* *Remote*
Built and tested data-integration services between enterprise systems and Oracle E-Business Suite.
- Built Java Kafka consumers that moved data between enterprise databases, deployed through Jenkins and OpenShift CI/CD pipelines.
- Created Splunk dashboards tracking the 200–400 users shared between enterprise systems and Oracle EBS, with alerts for

production incidents; prepared and validated Oracle EBS data with SQL.
- Wrote full test suites for 5+ Python services in the CI pipeline, keeping production defect rates low.

**Postdoctoral Research Associate** **Feb 2021 – Apr 2022**
*Rochester Institute of Technology* *Rochester, NY*
HPC simulations of supermassive black hole binaries in an NSF-funded project.
- Led a 7-person team upgrading PatchworkMHD, a C code for magnetohydrodynamic simulations of supermassive black hole binaries, in an NSF-funded project.
- Added black hole spin to the simulations with under 5% runtime overhead.
- Ran and benchmarked large production simulations on Frontera (TACC), about 10,000 node-hours, and identified performance bottlenecks in the parallel code.
- Calibrated Paramo model parameters against Fermi-LAT gamma-ray spectra with gradient-based optimization, and supervised a graduate student applying gradient descent to the same problem.
- Co-authored 2 papers from a multi-institution study of binary neutron star mergers [6, 7] and a third on blazar heating and cooling [8].

**Postdoctoral Research Fellow** **Oct 2018 – Nov 2020**
*Purdue University* *West Lafayette, IN*
Radiative-transfer modeling of relativistic jets (blazars and gamma-ray burst afterglows).
- Architected Paramo, an open-source Fortran 95 library that solves the Fokker–Planck equation coupled to radiative transfer; parallelized and optimized it with OpenMP, making it about 20× faster than the original serial code on 5 cores.
- Paramo has been adopted by other research groups, including Combi & Siegel (2023).
- Co-I on two NASA Fermi Guest Investigator proposals; one was funded ($68K) and supported the finding that blazar jets are launched with similar energy per baryon, independently of their power [5].
- Added numerical-integration and OpenMP features to Paramo and built Python tools to compute energy losses and spectra of high-energy particles in gamma-ray burst afterglows [4].
- Mentored 3 graduate students (1 M.S., 2 Ph.D.), co-authoring papers with each.
- Modeled the COVID-19 outbreak in Mexico (Python) and co-wrote Spanish-language infographics in a campaign against misinformation.

**Postdoctoral Research Fellow** **Jan 2018 – Sep 2018**
*Universidad Michoacana de San Nicolás de Hidalgo* *Morelia, Mexico*
- Built a Python pipeline that turned GRTrans black hole simulation images into training data for an SVM classifier of black hole spin (grayscale conversion, intensity normalization, spatial feature extraction).
- Released an open-source Python tool for radiative-transfer spectra and light curves.
- Organized an HDF5 workshop to train graduate students in high-volume data storage.

**Graduate Research Assistant** **Oct 2011 – Jul 2017**
*Universitat de València* *Valencia, Spain*
- Automated runs of the C-SPEV radiative-transfer code with Shell and Python pipelines: 10–60 simulations per project and tens of GB of HDF5 output, with data-quality checks for downstream analysis.
- Analyzed Fermi-LAT and VLBA multi-wavelength observations, cross-matching sources across catalogs and fitting log-parabola spectra.
- Fit C-SPEV simulation models to telescope data, identifying spectra and light-curve patterns that quantify plasma magnetization in blazars.
- Extended C-SPEV to compute discrete and continuous spectra from particle distributions of arbitrary shape, without affecting simulation runtime.
- Published 2 first-author papers in MNRAS [2, 3].

# Education

**Ph.D. in Physics** **Oct 2011 – Jul 2017**
*Universitat de València* *Valencia, Spain*
Excellent *cum laude*
Advisors: Prof. Miguel A. Aloy and Dr. Petar Mimica
Thesis: Numerical treatment of radiation processes in the internal shocks of magnetized relativistic outflows

**M.Sc. in Physics** **Aug 2009 – Sep 2011**
*Universidad Michoacana de San Nicolás de Hidalgo* *Morelia, Mexico*
Advisor: Prof. José A. Cervera
Thesis: *Study of TOV stars with the SPH method*

**B.Sc. in Physics** **Aug 2004 – Dec 2008**
*Universidad Autónoma del Estado de México* *Toluca, Mexico*
Advisor: Prof. Francisco S. Guzmán
Thesis: *Numerical solution of null geodesics for the generation of gravitational lenses produced by spherically-symmetric and static spacetimes*

Lic. Juan Josafat Pichardo Cruz Award

# Publications

## Articles

[12] Winkowski, S., **Rueda-Becerril, J. M.**, & Willing, C., *Using Numerical Simulations of Wind on Hilly Terrain to Model Propagule Dispersal*, in prep.

[11] Holzman, L., **Rueda-Becerril, J. M.**, Creamer, C., & Willing, C., *Random Forest Link Landscape Position to Drought Resilience in Giant Sequoia Forests*, in prep.

[10] **Rueda-Becerril, J. M.**, & Winkowski, S., & Willing, C. *WindsOfChange: A New Toolkit for Modeling Spore Dispersal in Hilly Terrains*, in prep.

[9] Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Tleco: A Toolkit for Modeling Radiative Signatures from Relativistic Outflows*, ApJ **976**, 182, (2024), arXiv:2405.17581.

[8] Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Balancing Turbulent Heating with Radiative Cooling in Blazars*, MNRAS **513**, 5766–5779, (2022), arXiv:2201.07790.

[7] Lopez-Armengol, F. G., Etienne, Z. B., [. . . ], **Rueda-Becerril, J. M.**, [. . . ] *Handing off the outcome of binary neutron star mergers for accurate and long-term postmerger simulations*, Phys. Rev. D **106**, 083015, (2022), arXiv:2112.09817

[6] Murguia-Berthier, A., Noble, S., [. . . ], **Rueda-Becerril, J. M.**, [. . . ] *HARM3D+NUC: A New Method for Simulating the Post-merger Phase of Binary Neutron Star Mergers with GRMHD, Tabulated EOS, and Neutrino Leakage*, ApJ **919**, 95, (2021), arXiv:2106.05356

[5] **Rueda-Becerril, J. M.**, Harrison, A. O. & Giannios, D. *Blazar jets launched with similar energy per baryon, independently of their power*, MNRAS **501**, 4092–4102, (2021), arXiv:2009.02273.

[4] Zhang, H., Christie, I., Petropoulou, M., **Rueda-Becerril, J. M.** & Giannios, D. *Inverse Compton Signatures of Gamma-Ray Burst Afterglows*, MNRAS **496**, 974–986, (2020), arXiv:1910.14049.

[3] **Rueda-Becerril, J. M.**, Mimica, P. & Aloy, M. A. *On the influence of a hybrid thermal–non-thermal distribution in the internal shocks model for blazars*, MNRAS **468**, 1169–1182, (2017), arXiv:1612.06383.

[2] **Rueda-Becerril, J. M.**, Mimica, P. & Aloy, M. A. *The influence of the magnetic field on the spectral properties of blazars*, MNRAS **438**, 1856–1869 (2014), arXiv:1310.5441.

[1] Guzmán, F. S. & **Rueda-Becerril, J. M.** *Spherical boson stars as black hole mimickers*, Phys. Rev. D **80**, 084023 (2009), arXiv:1009.1250.

## Proceedings

5. **Rueda-Becerril, J. M.** *A numerical approach for radiative cooling in relativistic outflows*, Astron. Nachr., 9th International Workshop on Astronomy and Relativistic Astrophysics: from Quarks to Cosmos **342**, 277–282, (2021), arXiv:2011.13797.

4. **Rueda-Becerril, J. M.**, Harrison, A. O. & Giannios, D. *The blazar sequence revised*, Astron. Nachr., 9th International Workshop on Astronomy and Relativistic Astrophysics: from Quarks to Cosmos **342**, 147–152, (2021), arXiv:2011.13805.

3. **Rueda-Becerril, J. M.**, Mimica, P. & Aloy, M. A. *Numerical simulations of the internal shock model in magnetized relativistic jets of blazars*, PoS(SWIFT 10) **233**, 159 (2014), arXiv:1502.07882.

2. **Rueda-Becerril, J. M.**, Mimica, P., Aloy, M. A. & Aloy, C. *Numerical study of broadband spectra caused by internal shocks in magnetized relativistic jets of blazars*, EPJ Web Conf. **61**, 02007 (2013), arXiv:1309.4612.

1. Mimica, P., Aloy, M. A., **Rueda-Becerril, J. M.**, Tabik, S. & Aloy, C. *Numerical simulations of dynamics and emission from relativistic astrophysical jets*, J. Phys.: Conf. Ser **454**, 012001 (2013), arXiv:1211.1794.

## Grants and Fellowships

### Research Grants

**NASA Fermi Guest Investigator Program, Cycle 12** **2019**
*A simple model to understand the blazar sequence* *Grant #121077, $68K*
PI: D. Giannios; Co-I: J. M. Rueda-Becerril

### Fellowships

**International Postdoctoral Fellowship** **Oct 2018 – Nov 2020**
*National Council of Science and Technology (CONACyT), Mexico*

**Postdoctoral Fellowship** **Jan 2018 – Sep 2018**
*Secretariat of Public Education, Mexico (Program for the Professional Development of Higher Education Institutions)*

**Study-Abroad Fellowship** **Sep 2014 – Aug 2016**
*National Council of Science and Technology (CONACyT), Mexico*

**Santiago Grisolía Fellowship** **Oct 2011 – Jun 2014**
*Council of Education, Research, Culture and Sport, Valencian Community, Spain*

**M.Sc. Fellowship** **Sep 2009 – Aug 2011**
*CONACyT, at the Institute of Physics and Mathematics, Universidad Michoacana de San Nicolás de Hidalgo*

**Summer Research Fellowship (XVII Summer of Scientific Research)** **Jun 2007 – Aug 2007**
*Mexican Academy of Sciences*

## Software and Projects

**Blazar Population ML** **Jan 2026 – Present**
*Physics-informed ML (Python, scikit-learn, PyTorch)* *github.com/altjerue/ilwikak-ml*
- Applying classical ML and Bayesian inference to the Fermi-LAT 4LAC-DR3 catalog (3,407 sources) to recover physical jet parameters from multi-wavelength data.
- Training a neural-network emulator of the physics-based Tleco model that maps physical jet parameters to observable spectral energy distributions, validated against Tleco output.
- Forecasting blazar gamma-ray flares with a time-series foundation model fine-tuned on Tleco-generated synthetic light curves.

**WindsOfChange** **Aug 2024 – Present**
*Lagrangian particle-dispersion library (C++/R)* *github.com/UW-ForestMycobiomeLab/WindSporesHills*
- Simulates particle dispersion, atmospheric turbulence, and flow over hilly terrain with forest canopy, based on analytical models (Massman & Weil 1999; Finnigan & Belcher 2004); designed to scale to tens of thousands of particles over kilometer-scale domains.

**Tleco** **Jan 2024 – Sep 2024**
*Radiative-transfer toolkit (Rust/Python)* *github.com/zkdavis/Tleco*
- Numerical PDE solvers, particle-distribution algorithms, and spectral calculations for relativistic plasmas; successor to Paramo [9].

**Paramo** **Oct 2018 – Apr 2022**
*Radiative-transfer library (Fortran 95, OpenMP)* *github.com/altjerue/paramo*
- Fokker–Planck and radiative-transfer solver for relativistic particle distributions; about 20× faster than the original serial code on 5 cores after OpenMP parallelization and optimization; used by other groups, including Combi & Siegel (2023).

**PatchworkMHD** **Feb 2021 – Apr 2022**
*Large-scale MHD simulation code (NSF project)*
- Led a 7-person upgrade of this production C code for HPC environments, adding black hole spin with under 5% runtime overhead.

## Awards

**Marcos Moshinsky Award for Best Poster** **Sep 2020**
*9th International Workshop on Astronomy and Relativistic Astrophysics (IWARA 2020), video conference*

**Lic. Juan Josafat Pichardo Cruz Award** **2009**
*Universidad Autónoma del Estado de México*
For defending a licentiate thesis within a year of completing the undergraduate coursework.

## Presentations

### Invited Talks

**The Role of Machine Learning in Environmental Science** **Mar 2024**
*Seminar on Climate Change Ecology, University of Washington* *Seattle, WA*

**Morphology of the spectra from numerical simulations of the internal shocks model for blazars** — **Feb 2019**
*Astrophysics Seminar, Purdue University* — *West Lafayette, IN*

**Numerical simulations of the internal shocks model in magnetized relativistic jets of blazars** — **Jun 2018**
*DATA Group Weekly Seminar, Instituto de Astronomía, UNAM* — *Mexico City, Mexico*

**Numerical treatment of non-thermal radiation in the internal shocks model for blazars** — **Mar 2018**
*Weekly Seminar, Instituto de Física y Matemáticas, Universidad Michoacana de San Nicolás de Hidalgo* — *Morelia, Mexico*

**Numerical simulations of the internal shock model in magnetized relativistic jets of blazars** — **Oct 2014**
*IVICFA's Fridays: Computation in Physics, IFIC* — *Paterna, Spain*

### Contributed Talks

**Simulations of supermassive binary black holes accretion dynamics in the spinning case** — **Apr 2022**
*APS April Meeting 2022* — *New York, NY*

**A numerical approach to the Klein-Nishina corrections of radiative cooling in relativistic outflows** — **Apr 2021**
*APS April Meeting 2021* — *Online*

**The blazar sequence revised** — **Sep 2020**
*9th International Workshop on Astronomy and Relativistic Astrophysics* — *Online*
Video: youtube.com/watch?v=BAZNWLNT69M

**Influence of the magnetic field on the spectral properties of blazars in the internal shocks scenario** — **Jun 2014**
*Extreme-Astrophysics in an Ever-Changing Universe* — *Ierápetra, Greece*

**Numerical study of broadband spectra caused by internal shocks in magnetized relativistic jets** — **Jul 2013**
*XXXIV Biennial Meeting of the Royal Spanish Society of Physics* — *Valencia, Spain*

### Posters

**A numerical approach for radiative cooling in relativistic outflows** — **Sep 2020**
*9th International Workshop on Astronomy and Relativistic Astrophysics* — *Online*
Marcos Moshinsky Award for Best Poster. Video: youtube.com/watch?v=0TJiKg7kOPI

**Numerical simulations of the internal shock model in magnetized relativistic jets of blazars** — **Dec 2014**
*Swift: 10 Years of Discovery* — *Rome, Italy*

**Numerical study of broadband spectra caused by internal shocks in magnetized relativistic jets** — **Jun 2013**
*The Innermost Regions of Relativistic Jets and Their Magnetic Fields* — *Granada, Spain*

## Teaching and Mentoring

**Zachary Davis, graduate student** — **2018 – 2022**
*Department of Physics and Astronomy, Purdue University* — *Mentoring*
Co-authored papers [8, 9]

**Amanda O. Harrison, graduate student** — **2018 – 2020**
*Department of Physics and Astronomy, Purdue University* — *Mentoring*
Co-authored paper [5]

**Hao Zhang, graduate student** — **2018 – 2019**
*Department of Physics and Astronomy, Purdue University* — *Mentoring*
Co-authored paper [4]

**Thermodynamics (graduate level)** — **Jun 2018**
*Instituto de Física y Matemáticas, Universidad Michoacana de San Nicolás de Hidalgo* — *Substitute Instructor*

## Professional Development

**High Performance Computing on Frontera** — **May – Jun 2021**
*Jason Allison et al., TACC, Austin, TX* — *Lecture series*

**Writing Winning Grants** — **Nov 2019**
*Dr. Lauren Broyles, Purdue University* — *Lecture*

**XSEDE HPC Workshop: Summer Boot Camp** — **Jun 2019**
*John Urbanic, Purdue University* — *Workshop*

**Data Analysis and Machine Learning with Python** — **Feb 2017**
*Dr. Alejandro Torres, Universitat de València* — *Workshop*

**Numerical Relativity Simulations of BBH Coalescence Using the Einstein Toolkit** — **Jul 2016**
*Dr. Vassilios Mewes, Universitat de València* — *Workshop, 8 hours*

**The Universe in the Light of PLANCK and BICEP2** — **May 2014**
*Prof. Nick Mavromatos, Universitat de València* — *Lecture series, 2 credits*

**Dark Matter** — **Sep 2013**
*Prof. Alejandro Ibarra, Universitat de València* — *Lecture series, 2 credits*

**International Cargèse School on Cosmic Accelerators** — **Apr – May 2013**
*Institut d'Études Scientifiques de Cargèse, France* — *Summer school*

**Introduction to C++ Programming** — **Apr 2012**
*Dr. Jacek Generowicz, Universitat de València* — *Workshop, 6 credits*

**Numerical Relativistic Astrophysics** — **Mar – Apr 2012**
*Prof. Luciano Rezzolla, Universitat de València* — *Lecture series, 9 hours*

**Fortran for Scientific Computing** — **Mar 2012**
*HLRS, University of Stuttgart, Germany* — *Workshop, 33 hours*

## Outreach

**Los más rápidos y los más furiosos (The Fastest and the Most Furious)** — **Sep 2020**
*Community of Undergraduate Physics Students, Juárez Autonomous University of Tabasco* — *Online talk*

**Una simulación de la física y la astrofísica (A Simulation of Physics and Astrophysics)** — **Aug 2020**
*Community of Undergraduate Physics Students, Juárez Autonomous University of Tabasco* — *Online talk*

**ANITA y la teoría de los universos paralelos (ANITA and the Theory of Parallel Universes)** — **May 2020**
*Científicos Mexicanos en el Extranjero* — *Blog post*
mexiciencia.github.io/post/anita

**¿Qué es el modelo SIR? (What is the SIR Model?)** — **May 2020**
*Científicos Mexicanos en el Extranjero* — *Blog post*
mexiciencia.github.io/post/modelo-sir

**Evolución del brote epidémico de COVID-19 (Evolution of the COVID-19 Epidemic Outbreak)** — **Apr 2020**
*Científicos Mexicanos en el Extranjero* — *Blog post*
Data analysis and modeling collaborator. mexiciencia.github.io/post/covid19

**Annual Department of Physics and Astronomy Poster Event** — **Nov 2019**
*Purdue University* — *3 posters*

**Post-Doc Panel Q&A: What Happens When We Complete Our PhDs?** — **Apr 2019**
*Department of Physics and Astronomy, Purdue University* — *Panelist*

**Annual Department of Physics and Astronomy Poster Event** — **Nov 2018**
*Purdue University* — *Poster*

**¿Decía Einstein la verdad? (Was Einstein Telling the Truth?)** — **Mar 2009**
*Facultad de Ciencias, Universidad Autónoma del Estado de México* — *Talk*

## Service

**Organizing contributor** — **Dec 2012**
*X Scientific Meeting of the Spanish Astronomical Society* — *Valencia, Spain*

**Student representative, Governing Council of the Faculty of Sciences** — **Aug 2007 – May 2009**
*Universidad Autónoma del Estado de México* — *Toluca, Mexico*

## Languages

Spanish (native), English (full professional proficiency), Catalan (basic), Nahuatl (basic)